When Identity Is Load-Bearing: A Proposed Design Law for Incorporated Developmental State

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Abstract

An incorporated developmental state, a bearer that both biases behavior and is updated by that behavior's own outcomes, is central to Bearer Theory's account of when a system has an identity that is its own. We ask when that state actually does causal work rather than sitting idle, and we propose a design law, supported by cross-substrate evidence, that predicts three regimes. When the task admits a non-incorporated route to viability, the bearer is idle, and increasing capacity increases the escape rather than the incorporation. When the route is foreclosed but the bearer cannot adapt to carry the required state, the system fails, the difficulty valley. When the route is foreclosed and the bearer is adaptive, the incorporated state becomes load-bearing and bearer-specific. We give operational definitions of these terms and show the mechanism across three substrates that share nothing but the loop: a reinforcement-learning forager, a transformer with a fast-weight memory, and a competitive-fatigue dynamical system. The loop and delayed incorporation appear in all three; the escape principle in two; the difficulty valley in two; and load-bearing non-transferability in the substrates that foreclose the escape with an adaptive bearer, where a self versus wrong-bearer gap of 0.51, replicated with tight error bars, dissociated from attention, and holding at scale, establishes it, and where the same result reproduces under gradient descent in a language model, recall of a private fact set being perfect with the bearer and at chance without it, on both an abstract key-value map and natural-language persona documents over real byte-pair tokens. A counterintuitive corollary follows for scaling: a more capable optimizer does not discover its bearer identity, it exploits the escape better. The claim is structural and agnostic on the phenomenal question by design.

Keywords: developmental state, fast weights, meta-learning, personal identity, substrate independence, incorporated memory, non-transferability, machine consciousness

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1. Introduction

Bearer Theory decomposes an agent into four loci over one continuant: an actor whose action is conditioned on a bearer state, an outcome constituted relative to that bearer, an update of the bearer from its own outcomes, and the recurrence of all three over a single persisting thing. Its Prediction 2 is that self-coupling, updating the bearer on one's own bearer-relative outcomes, yields an advantage over a yoked control. The sharpened question is not whether a bearer can be built, but when it does causal work.

A single demonstration invites the objection that a bearer effect is an artifact of one architecture. The claim earns the word law only if it recurs in formalisms that share no implementation and were not built to agree. This paper reports the same structure in three: an evolved reinforcement-learning forager with outcome-gated fast weights, a small transformer whose incorporated state is a fast-weight associative memory read alongside attention, and a dynamical system in which tangent-force subsystems compete for control of a state on a manifold under fatigue. The forager and the transformer were built for this program. The dynamical system, FourD, was built independently for a different purpose and arrived at the bearer loop on its own. What transfers is not code. It is a loop and a law.

The core result, informally: identity does not become load-bearing because a system has memory. It becomes load-bearing when the system can no longer succeed by treating its history as somebody else's history, or as no history at all.

Contributions.

1. A proposed design law with three regimes (escape open, escape closed with a static bearer, escape closed with an adaptive bearer), stated formally with operational definitions.
2. Cross-substrate evidence for both sides of its mechanism: the loop and incorporation in all three substrates, the escape principle in two, the difficulty valley in two, and load-bearing non-transferability in the one that forecloses the escape adaptively.
3. A scaling corollary: capacity does not imply incorporation when an escape exists; it implies better exploitation of the escape.
4. Falsifiable predictions for new substrates and a concrete experiment that would convert the dynamical system from an escape demonstration into a positive one.

2. Related work

The bearer loop is a fast-weight architecture in the sense of Hinton and Plaut (1987) and of Schmidhuber's fast-weight programmers (Schmidhuber, 1992): a slow network sets, and a fast state carries, information that changes within a single episode. Our incorporated

memory is a learned-write fast weight, close in mechanism to modern fast-weight and linear-attention memories (Ba et al., 2016; Schlag et al., 2021), but the question we ask is different. That literature asks whether such a memory improves performance. We ask when it becomes the only viable carrier of a behavior and therefore non-transferable, which is a question about task structure, not memory capacity.

The self-versus-yoked comparison is the meta-learning setup of learning to reinforcement learn (Duan et al., 2016; Wang et al., 2016), in which a recurrent state adapts within an episode, a lineage that runs back through learning to learn by gradient descent (Hochreiter et al., 2001; Finn et al., 2017). Our result explains a failure mode of that setup: when the environment leaves a bearer-neutral route, the adapting state is idle, and a more capable meta-learner exploits the route more. This is a caution for reading emergent in-context adaptation as a sign of a persistent self.

The record-versus-bearer distinction is the episodic-memory-versus-weights distinction made sharp. Retrieval-augmented and long-context models treat history as a record that is read (Lewis et al., 2020), and such records have known limits as they lengthen (Liu et al., 2023); our foreclosure result shows that when the needed information cannot be held in a bounded context, an incorporated state carries it and a record cannot, framed as a structural necessity rather than an efficiency.

Philosophically the target is the developmental and constitutive component of personal identity, the Lockean intuition that a self is constituted by an appropriated history (Locke, 1689), rendered as a non-transferability property rather than a memory-continuity criterion. We part from psychological-continuity accounts (Parfit, 1984) by locating identity in an incorporated state that cannot be copied into another actor without failing, and we take no position on the phenomenal question (Nagel, 1974), keeping company with skeptical treatments of near-future machine consciousness (Schwitzgebel, 2026).

3. The bearer loop and the four loci

Write the loop as

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a_t = pi(b_t, w_t)        act, conditioned on the bearer
o_t = V(e_t, b_t)          outcome, constituted relative to the bearer
b_{t+1} = U(b_t, o_t)      update the bearer from its own outcome
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over one continuant b. Each substrate instantiates all four loci: the forager (bearer plus outcome-gated fast weights, a recurrent policy, a bearer-relative homeostatic outcome, a plastic update), the transformer (a fast-weight associative memory W, a bilinear read, a reward gated by the chosen arm, a reward-gated Hebbian write), and the dynamical system

(a persistent vector b_t biasing perception, subsystem activities, an outcome filtered through b_t, and b_{t+1} = (1 - decay) b_t + gain activities_t). The correspondence is exact, not analogical: FourD's own words, a closed loop and not a passive log, are Prediction 4.

4. Definitions and the proposed design law

Let J be a task-viability score, tau a viable threshold, and B a bearer under a fixed actor and update rule.

- **Load-bearing.** B is load-bearing on task T if freezing its update costs viability:

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J(B_self) − J(B_frozen) > epsilon.
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- **Bearer-specific (non-transferable).** B is bearer-specific if updating it on another individual's outcomes, matched in volume, fails on one's own criterion:

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J(B_self) − J(B_wrong) > epsilon,   with J(B_self) − J(B_falsifier) ~ 0,
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where the falsifier is the same individual's outcomes with independent noise.

- **Escape route.** An escape exists for T if some policy pi_g that does not require incorporated bearer history reaches viability:

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E(T, tau):   there exists pi_g with J(pi_g) >= tau.
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- **Foreclosure.** T forecloses the escape if no such policy exists:

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not E(T, tau):   there is no pi_g with J(pi_g) >= tau.
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- **Adaptivity.** A(B) holds if the bearer can acquire, from its own outcome history, the task-specific state that viability requires.

Proposed design law.

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LB(B, T)  and bearer-specific(B, T)   <=>   ( not E(T, tau) )  and  A(B).
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In words: a developmental bearer becomes load-bearing and bearer-specific when task viability requires information that can only be acquired through that bearer's own incorporated outcome history, and the bearer is capable of adapting to carry it. If a non-incorporated viable policy remains, optimization preferentially exploits that route instead.

The law separates into three regimes, which the substrates instantiate (Figure 1):

Task structure	Bearer	Result
escape open (E)	any	bearer idle
escape closed (not E)	inadequate or static	difficulty valley (system fails)
escape closed (not E)	adaptive (A)	incorporated state becomes load-bearing

We do not claim to have proven the universal biconditional. We claim cross-substrate evidence for both directions of the mechanism, and we state below the experiment that would occupy the presently thin cell (escape closed, adaptive) in a second substrate.

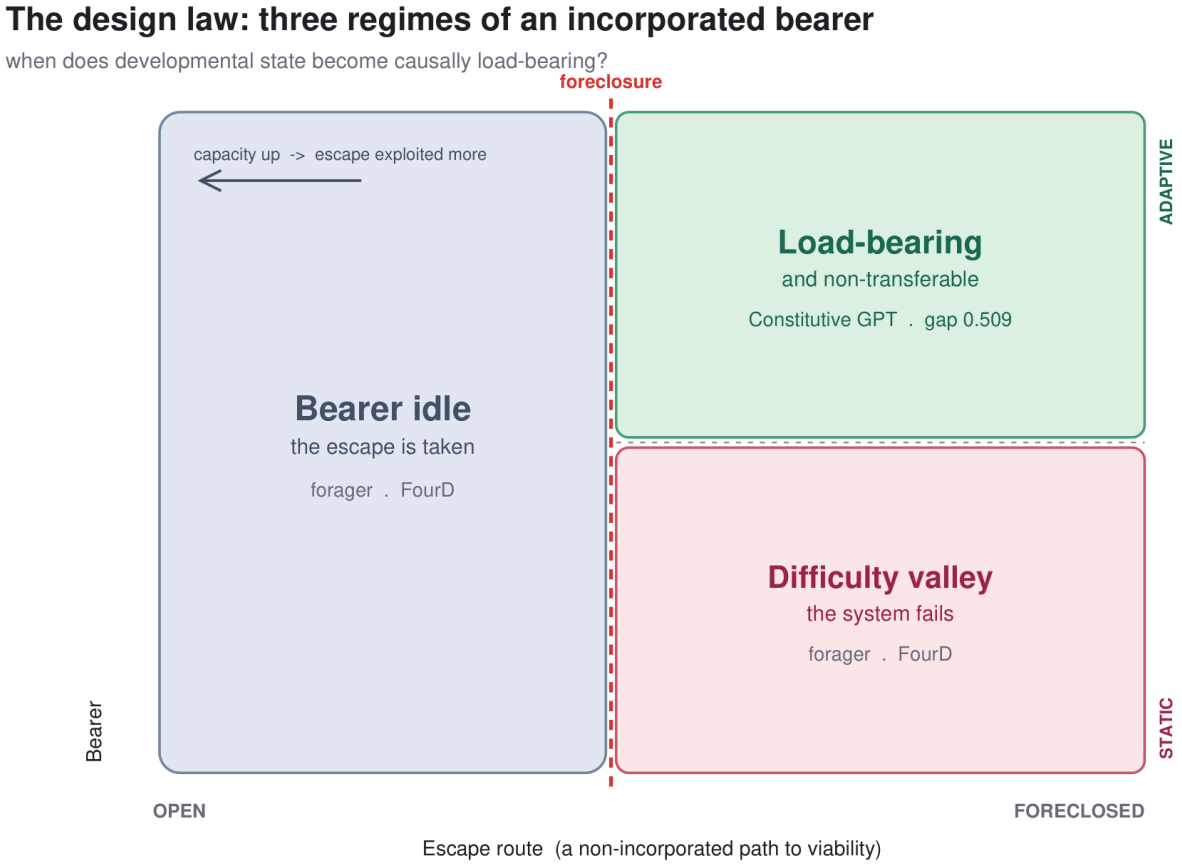


Figure 1: The design law as a 2D design space, escape (open to foreclosed) against bearer adaptivity (static to adaptive). The bearer is load-bearing and non-transferable only in the top-right cell, where the escape is foreclosed and the bearer is adaptive. With an escape open, capacity only deepens its exploitation.

5. The loop and incorporation are substrate-independent

Delayed incorporation, a specific outcome leaving a residue that persists after its cue, is measured in all three substrates: the forager (a delayed incorporation magnitude of 5.76 in the continuant probe), the transformer (the memory integrates its own outcomes over a horizon a bounded context cannot hold; freezing the write removes exactly that), and the dynamical system (a one-shot perturbation persists about 120 steps, a memory horizon that is 0 without the bearer). The loop and its incorporation are not features of one architecture.

6. The escape principle is substrate-independent, and a scaling corollary

The bearer stays idle whenever the task leaves a non-incorporated route to viability.

- **Forager.** Across nine task designs and two capacity levers, self-coupling produced no bearer-neutral advantage, because each task retained a bearer-independent viable policy.
- **Dynamical system.** A clean null: transplanting a different body's incorporated bearer into body A, matched in volume, changes A's competency within noise (self-maintenance 2.714 versus 2.716, basin divergence 0.003 versus 0.003 against a same-body falsifier), because A's fixed subsystem preferences are a non-incorporated route and the bearer is a weak add-on the dynamics wash out.

The scaling corollary. The two forager levers make the sharp point. Softening the viability threshold re-admitted a quantity route, and giving the learner a richer update rule made it exploit that route more efficiently, not the bearer.

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capacity up, escape present  =>  escape exploitation up,  not  incorporation up.
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A more capable optimizer does not discover its bearer identity; it finds the escape better. This is a design caution: increasing model capability, context length, retrieval quality, or agent-loop sophistication is not evidence that a persistent developmental identity will emerge. A sufficiently capable system may become better at avoiding the very channel one wanted to make constitutive.

7. Foreclosure with an adaptive bearer yields load-bearing non-transferability

The transformer is the one substrate that forecloses the escape by construction and has an adaptive bearer. Its task is stochastic bandit identification: a hidden per-instance code, weakly informative outcomes, so identifying the code requires integrating outcomes over a horizon far longer than any bounded context. Gate zero is an information bound, computed with no training: a context-only learner restricted to the last L outcomes stays near chance (0.170 at the locked calibration) while the full horizon is solvable (0.874). The context window, the sequence model's most powerful escape, is foreclosed.

On that task the incorporated memory is load-bearing and bearer-specific by the definitions of section 4.

- **Load-bearing.** J(B_self) - J(B_frozen) = 0.607: freezing the write collapses accuracy to exact chance. A dual-channel model with a real attention block gives the double dissociation, removing the attention record drops accuracy by 0.003 (record idle) while freezing the memory collapses it to the context bound (bearer carries the task).
- **Bearer-specific.** J(B_self) - J(B_wrong) = 0.509 plus or minus 0.049 across six seeds, wrong-bearer at exact chance, the falsifier holding and isolating the effect. It is not a one-bit lookup: a multi-dimensional vector identity shows the same non-transferability. And it holds at scale, reproducing at 133,637 parameters with a genuine transformer block, the gap if anything larger.

The same result under gradient descent, in a language model. The foregoing was trained by evolutionary search on an abstract task. The positive cell also holds under gradient descent on token sequences in a real decoder transformer (Figure 2). The task is associative recall of a private per-sequence key-to-value map: study pairs, then a filler that pushes the study block outside a windowed attention of size L, then queries. The bounded window is the record and cannot reach the study block; a delta-rule fast-weight memory, a fixed-size matrix written online, is the bearer. Trained end to end with cross-entropy, the transformer recalls the map perfectly (query accuracy 1.000). Ablating the bearer, by zeroing its read, drops accuracy to chance (0.064); reading one sequence's queries against a different sequence's bearer likewise drops to chance (0.056). Load-bearing and non-transferability, at near-perfect separation, in the substrate the term language model names. The escape had to be foreclosed here too: a bijection map leaked a process-of-elimination route within the window and left the bearer idle, and a random-with-replacement map closed it, the same lesson a fourth time.

And on natural language. The same architecture holds when the task is rendered in English over real byte-pair tokens rather than abstract symbols (Figure 2, right). Each document is a private persona: a block of natural-language facts (a favorite color, a city, a pet, a job), then a stretch of ordinary filler sentences that pushes the fact block outside the attention window, then natural questions ("Q: What is Maria's favorite color? A:"), all tokenized with the GPT-2 byte-pair encoder. Trained end to end, the transformer recalls the persona's facts perfectly (answer accuracy 1.000); ablating the delta-rule bearer drops it to chance (0.129, chance 0.100) and answering one document's questions against a different persona's accumulated bearer drops it to chance (0.094). Load-bearing gap +0.871, non-transferable gap +0.906, in the substrate of ordinary text. Here the escape needed no

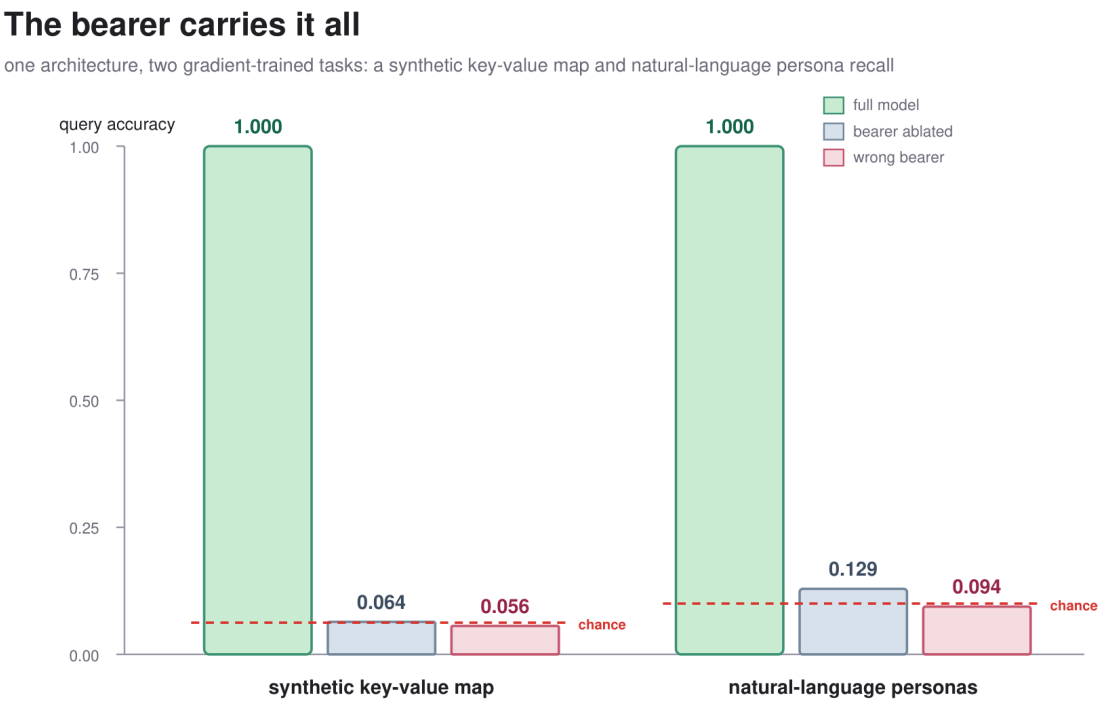


Fig. 2. The design law's positive cell under gradient descent, in two decoder transformers with a delta-rule fast-weight bearer. Both recall a private per-sequence fact set perfectly; ablating the bearer's read or reading against another sequence's bearer collapses accuracy to chance. Left: an abstract key-value map (chance 1/16). Right: natural-language personas over real GPT-2 BPE text (chance 1/10). Load-bearing and non-transferable in both, in the substrate the term language model names.

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special foreclosure: natural filler carries no systematic cue to the facts and the window cannot reach them, so the incorporated bearer is the only route by construction, where the synthetic bijection had leaked one. The positive cell thus holds under gradient descent on both abstract tokens and natural language.

Robustness and scale of the natural-language cell. The effect is neither a lucky seed nor a single operating point. Across five seeds the load-bearing gap is 0.915 +/- 0.026 (95% CI [0.892, 0.938]) and the non-transferable gap 0.960 +/- 0.005 ([0.955, 0.964]), with every seed a pass. The bearer behaves as a finite store with a real capacity: at a load of six facts a bearer of dimension eight cannot hold them (recall 0.48) while dimension sixteen and up saturates near ceiling, and at fixed size the same bearer carries a growing fact set (recall 1.00, 1.00, 0.93 at two, four, and six facts). And the cell survives long documents: with a chunk-parallel delta rule, numerically identical to the sequential scan and about ten times faster at 768 tokens, the bearer stays load-bearing (+0.68) and non-transferable (+0.67) across a context twice as long, where recall is bounded by memory size against facts times context length, the same capacity ceiling reached from the length axis rather than any loss of the load-bearing structure.

8. The difficulty valley

Foreclosure is necessary but not sufficient, which is the middle row of the regime table. On a substrate whose bearer is not adaptive, foreclosing the escape leaves the identity unsustainable. In the forager, an obligate task with the escape closed became unlearnable at the smallest scales. In the dynamical system, a foreclosure attempt that made identity live only in the bearer produced the effect at short horizon (basin divergence 0.103 under the wrong bearer against 0.008 for the falsifier) but lost it at the full horizon (0.004 versus 0.004): a seeded identity is transient, and without an ongoing objective the loop converges to a generic attractor and the identity washes out. The lesson is the same in both, and it is why the law contains adaptivity: a bearer must integrate a body-specific parameter from its own outcomes that behavior then requires. The transformer alone had both, which is why it alone reaches the positive result. The valley explains why prior attempts to force persistent identity yielded either nothing (escape open) or a broken learner (escape closed, static): the middle variable, adaptivity, was missing.

9. Cross-substrate summary

	forager (RL)	transformer (GPT)

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bearer loop (four loci)                 yes                          yes
incorporation (Prediction 3)
> 120
escape principle (idle)                 yes, 9 designs + 2 levers    n/
a (foreclosed)          yes, self ~= wrong null
difficulty valley                       yes (obligate unlearnable)   avoided
foreclosure + adaptive -> load-bearing  not reached                  yes, freeze -
> chance     not reached
bearer-specific (non-transferable)      not reached                  yes, gap 0.509 +/-
 0.049  not reached
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The transformer column holds three ways. Under evolutionary search on the abstract task it gives the numbers above; under gradient descent on abstract token sequences (the demo language model, section 7 and Figure 2, left) it gives perfect recall with the bearer and chance without it; and under gradient descent on natural-language persona documents over real GPT-2 byte-pair tokens (Figure 2, right) it gives the same, load-bearing $+0.871$ and non-transferable $+0.906$. The positive cell is thus established across both training regimes and on ordinary text.

10. Falsifiable predictions

- Any substrate with a surviving non-incorporated route shows an idle bearer, a null self versus wrong-bearer, regardless of capacity.
- Foreclosing that route, and only that, together with an adaptive bearer, makes the bearer non-transferable.
- The concrete stated test is the FourD adaptive foreclosure: give the dynamical system's bearer a task-driven objective that requires an integrated body parameter, then repeat the self versus wrong-bearer test. The prediction is that non-transferability then appears.

11. Discussion, limitations, and the three-paper program

The tasks are deliberately minimal so that the causal structure is legible; that is a strength for a law and a limit for claims about rich behavior. The positive result now holds under evolutionary search at 133,637 parameters, under gradient descent in an abstract token-sequence transformer with a delta-rule memory, and under gradient descent on natural-language persona documents over real GPT-2 byte-pair tokens (Figure 2), where the foreclosed escape is the context window and the natural filler closes it by construction. The remaining step toward scale is a larger model trained on a general corpus, where the obligate task is drawn from the text itself rather than constructed. The dynamical system remains an escape demonstration until its bearer is made adaptive.

Evolutionary search was noisy at the larger sizes; the self-versus-wrong result is reported with across-seed error bars for that reason. The law is about causal load and

transferability, not experience.

This paper sits in the middle of a three-layer program, and the layers should be read as distinct: Bearer Theory (Bone, 2026a) supplies the formal object and the developmental loop; this paper supplies the empirical design law governing when that object becomes causally necessary; and Life Before Language (Bone, 2026b) supplies the philosophical interpretation of how far such engineering reaches toward consequence and subjectivity. Keeping the empirical layer free of the phenomenal question is what makes it hard to dismiss as metaphysics dressed as machine learning: here is a measurable property, bearer-specific developmental dependence, and its governing conditions; what it means is a separate question, asked elsewhere.

12. Honest scope

Randomness quality was tested in the dynamical system, a pseudo-random source against the operating system's cryptographic source under a two-one-sided-tests equivalence check, and was found not to drive the incorporation effect: perturbation shape does, not entropy source. A hardware quantum source was deliberately not claimed, since none was run, and an early apparent quantum-randomness effect was traced to a shared-RNG-stream artifact and corrected. Incorporation is a structural property of the loop, not of the noise.

13. Methods and reproducibility

The transformer code and results are public under MIT at github.com/vikingdude81/constitutive-gpt, with the milestone map and trained policies, and the forager foundation in the same repository. The entropy-decay probe is public at github.com/vikingdude81/gabriels-horn. The dynamical system is the FourD repository, with the bearer loop in bearer_state_competency.py and the self versus wrong-bearer and foreclosure tests in fourd_self_vs_wrong_bearer.py and fourd_foreclosure.py. The gradient-trained demo language model is in demo_lm.py. Training in the forager and the evolutionary-search transformer is evolutionary search; the demo language model is trained by gradient descent (AdamW, cross-entropy); the dynamical system is a fixed integrator; the gate-zero bound is a Monte Carlo of the optimal decoder and needs no training. Per-experiment numbers are in the RESULTS files of each repository.

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